Candidate's name._____**Sign** ____

Personal number _____

UACE CHEM P 2 **SET 10**

EDUCATOR'S FORUM

Uganda Advanced Certificate of Education

Uganda Advanced Certificate of Education

CHEMISTRY PRACTICAL (PAPER 2)

Time: 3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- This examination paper consists of two compulsory examination items.
- Responses to these items must be written in the answer booklet provided.
- Begin each item on a fresh page.
- All working must be clearly shown. Graph papers will be provided.

For examiners use only:

COMPETENCE	P	M	Dp	Da	O	C&R
ITEM 1						
ITEM 2						

ITEM 1

A local cottage industry in Western Uganda extracts essential oils and bio-active compounds from indigenous trees for pharmaceutical purposes. During a fractional distillation run, a technician isolated a liquid fraction code-named P1. The compound exhibits strong antioxidant activity and is suspected to be a phenolic derivative that could replace expensive imported synthetic preservatives. However, the production team requires absolute identification of its functional group family before scaling up production. You are provided with 15 cm^3 of liquid P1.

Task: Design and carry out a systematic qualitative analysis investigation to confirm the functional group family and carbon backbone saturation level of sample P1. Perform a minimum

of 6 distinct diagnostic tests . Based on your findings, write a suitable IUPAC name, draw its structural formula, and explain its reactive properties in standard laboratory environments.

ITEM 2

A domestic chemical enterprise produces a commercial bathroom descaler containing orthophosphoric acid (H3PO4) as the active ingredient. A quality assurance officer noted that a newly delivered batch of raw materials failed to remove lime scale efficiently. The officer suspects the batch contains unreactive soluble neutral salts that diluted the acid concentration. The factory requires a minimum purity of 85% by mass for production. A junior technician suggested weighing 2.50 g of the crude acid sample, reacting it with an excess of standard sodium hydroxide solution, and back-titrating the unreacted alkali. However, the technician struggled with the multi-step stoichiometric calculations required after dilution.

- **FA1:** 0.50 mol dm^{-3} sulphuric acid solution (for standardisation)
- **FA2:** Sodium hydroxide solution of unknown concentration
- **Solid X1:** 2.50 g of the crude orthophosphoric acid sample
- **Indicators:** Methyl orange and phenolphthalein indicators
- **Solvent:** Distilled water
- Relevant reactions
- Standardisation of Sodium Hydroxide FA2 using Sulphuric Acid
  FA1:

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

- Reaction of the Crude Acid Sample t{Solid } X1 with Excess Sodium Hydroxide

$$H_3PO_4(aq) + 3NaOH(aq) \rightarrow Na_3PO_4(aq) + 3H_2O(l)$$

Back-Titration of Unreacted (Excess) Sodium Hydroxide with Sulphuric Acid
FA1:

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Task

- Design and carry out a quantitative back-titration experiment to determine the percentage purity of the orthophosphoric acid sample in Solid X1. Present your titration data in a clear table showing at least six accurate titre values (standardisation and main titration sets), and show all steps of your calculations, including balanced equations for all reactions involved. Advise the enterprise whether the batch meets the minimum purity requirements.

END